

WORKSHEET - NAPLAN Year 9

Number and Algebra Solving Equations (9)

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Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number and Algebra, NAP-26179

David buys a surfboard cover online that costs him \$128.

The postage costs involved are listed in the table below.

Weight	Up to 500 g	500 g to less than 1 kg	1 kg to less than 1.25 kg	1.25 kg to less than 1.5 kg
Postage	\$12	\$17.50	\$21.75	\$25

The surfboard cover weighs 1.1 kg

What does David have to pay, in total, to purchase the surfboard and have it delivered?

\$140 \$145.50 \$149.75 \$153

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2. Measurement and Geometry, NAP-26177

A bullet fired from a hand gun travels at 390 metres per second.

What is the speed of the bullet in metres per minute?

39 000 23 400 6.5 3.9

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3. Number and Algebra, NAP-55641

A store sells cups for \$4 each and bowls for \$5 each.

Alex wrote the equation $4c + 5b = 65$ to calculate the number of cups and bowls he could buy for exactly \$65.

How many cups and bowls did Alex buy for \$65?

☐ **3** cups and **10** bowls

☐ **4** cups and **10** bowls

☐ **5** cups and **9** bowls

☐ **10** cups and **7** bowls

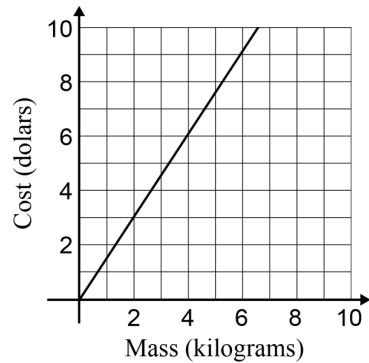
4. Number and Algebra, NAP-49728

Paddy and Miffy each bought potatoes from the local farmers' market.

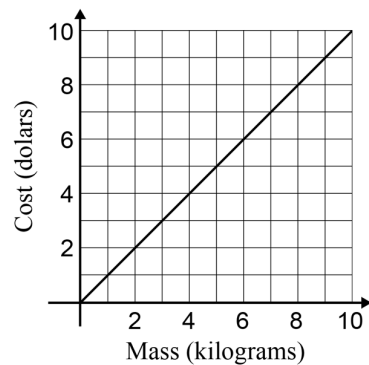
Paddy bought 2 kilograms for \$2.50.

Miffy bought 6 kilograms for \$7.50.

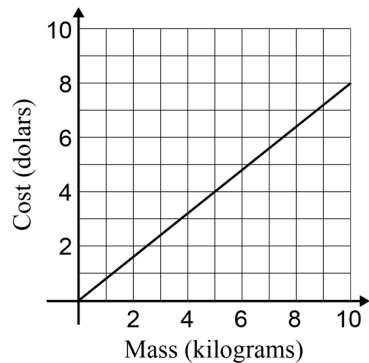
Which graph best represents the cost of potatoes at the farmers' market?



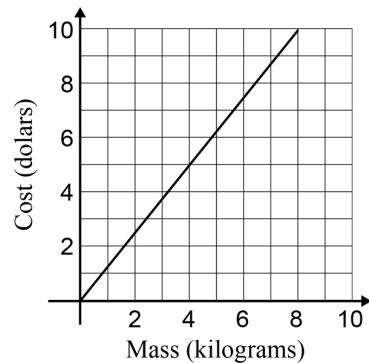
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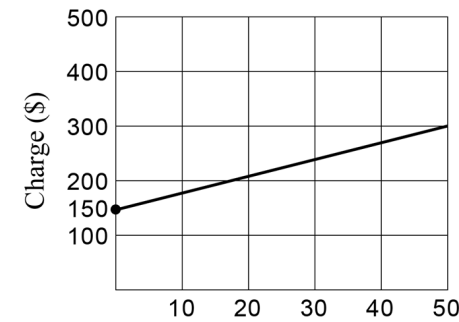
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5. Number and Algebra, NAP-67369

Bella is an electrician.

She charges \$150 to attend a job and then a fixed price for each minute she spends on the job.

The graph below shows Bella's charge based on the number of minutes she spends at the job.



How much will Bella charge for a job that takes 60 minutes?

\$

6. Number and Algebra, NAP-55702

Willy buys 16 lollipops for \$5.60.

Wonka buys 30 lollipops at the same cost per lollipop.

How much more does Wonka pay than Willy?

\$

7. Number and Algebra, NAP-26268

Cassie opens a savings account and deposits \$900 into it.

She makes no more deposits and earns simple interest on her original deposit at 3.5% each year.

How much interest will Cassie earn after 4 years?

\$

8. Number and Algebra, NAP-96834

In this inequality n is a whole number.

$$\frac{8}{n} < \frac{5}{8}$$

What is the smallest possible value for n to make this inequality true?

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Worked Solutions

1. Number and Algebra, NAP-26179

$$\begin{aligned} \text{Surfboard cover cost} + \text{postage} \\ &= \$128 + \$21.75 \\ &= \$149.75 \end{aligned}$$

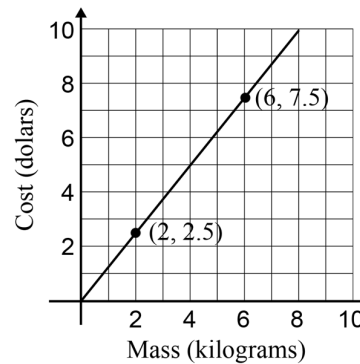
2. Measurement and Geometry, NAP-26177

$$\begin{aligned} \text{Speed} &= (390 \times 60) \\ &= 23\,400 \text{ metres per minute} \end{aligned}$$

3. Number and Algebra, NAP-55641

$$\begin{aligned} (4 \times 5) + (5 \times 9) &= 20 + 45 = \$65 \\ \therefore & \text{5 cups and 9 bowls.} \end{aligned}$$

4. Number and Algebra, NAP-49728



5. Number and Algebra, NAP-67369

Charge to attend = \$150

From the graph,

$\Rightarrow$ 50 minute job costs \$300

$$\therefore \text{Cost per minute} = \frac{300 - 150}{50} = \$3$$

$\therefore$ Cost of 60 minute job

$$= 150 + 60 \times 3$$

$$= \$330$$

6. Number and Algebra, NAP-55702

$$\text{Cost of 1 lollipop} = \frac{5.60}{16} = \$0.35$$

$$\text{Cost of 30 lollipops} = 30 \times 0.35 = \$10.50$$

$\therefore$ Extra that Wonka pays

$$= 10.50 - 5.60$$

$$= \$4.90$$

7. Number and Algebra, NAP-26268

Interest earned = Prn

$$= 900 \times \frac{3.5}{100} \times 4$$

$$= \$126$$

8. Number and Algebra, NAP-96834

$$\frac{8}{n} < \frac{5}{8}$$

$$5n > 64$$

$$n > \frac{64}{5}$$

$$> 12.8$$

$\therefore$ Smallest $n = 13$.